NEXT GEN
SOFTWARE HUB

CYBER SECURITY INTERNSHIP

TASK 02
TASK REPORT

Networking Fundamentals and Network Analysis



TOPICS

- OSI Model
- TCP/IP Model
- Firewalls & VPNs
- Network Devices
- IP Addresses & Packets
- Nmap & Wireshark

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1. **INTRODUCTION**

Computer networking is the process of connecting computers, servers, network devices, applications, and other systems so that they can communicate and exchange information. Networks allow users to access websites, communicate with servers, share resources, use applications, and access services. Networking is also an important part of cybersecurity. When systems communicate over a network, information is transmitted between different devices. Security professionals therefore need to understand how network communication works and how to identify network services and traffic.

This task focuses on networking fundamentals, including **the OSI Model, TCP/IP, firewalls, VPNs, network devices, IP addresses, network packets, Nmap, and Wireshark**.

For the practical part, Kali Linux was used as the analysis machine and DC-1 from VulnHub was used as the target machine. Netdiscover was used to identify the target IP address. Nmap was then used to identify open ports and services. After identifying HTTP on port 80, the DC-1 web server was accessed using Firefox. Wireshark was then used to capture and analyze the network traffic.

2. **OBJECTIVES**

The objectives of this task were:

- To understand the fundamentals of computer networking.
- To understand the OSI Model.
- To understand the TCP/IP model.
- To understand IP addresses and network packets.
- To understand TCP and UDP.
- To understand common network devices.
- To understand the purpose of firewalls.
- To understand the purpose of VPNs.
- To understand basic network security.
- To understand the purpose of Nmap.
- To understand the purpose of Wireshark.
- To discover a target machine using Netdiscover.
- To scan the target using Nmap.
- To identify open ports and services.
- To access an HTTP service.
- To capture network traffic using Wireshark.
- To analyze captured network packets.

3. **INTRODUCTION TO COMPUTER NETWORKING**

A computer network is a group of connected devices that communicate with each other.

Examples of network devices include:

- Computers
- Laptops
- Servers

- Routers
- Switches
- Firewalls
- Printers
- Smartphones
- IoT devices

Networks allow these devices to exchange information and share resources. For example, when a user enters a website address into a browser, the computer sends a request to a web server. The server processes the request and sends a response back to the user's computer.

Network communication uses different protocols and addressing systems to make sure that information reaches the correct destination.

Important networking concepts include:

- IP addresses
- MAC addresses
- Ports
- Protocols
- Packets
- Network devices

Understanding these concepts is important for cybersecurity because network traffic can provide information about systems, services, and communication between devices.

4. **ENTERPRISE NETWORKING**

Enterprise networking refers to the private network infrastructure used by organizations to connect their offices, employees, devices, data centers, cloud environments, and other resources.

An enterprise network may contain:

- Branch offices
- Internal data centers
- Local Area Networks (LANs)
- Wide Area Networks (WANs)
- Cloud environments
- Employee devices
- Servers
- IoT devices
- Network appliances

Enterprise networks allow different parts of an organization to communicate and share resources. Organizations can also use technologies such as VPNs and encryption to protect communication between users and internal resources.

Traditional enterprise networks often relied on centralized data centers, LANs, WANs, and technologies such as MPLS.

Modern enterprise networks increasingly use cloud services, remote access, SD-WAN, SASE, and other technologies because employees and applications may be distributed across different locations.

5. NETWORK DEVICES

Different network devices perform different functions.

5.1 Routers

A router connects different networks.

It forwards packets between networks using IP addresses and routing information.

For example, a router can connect a local network to the Internet.

5.2 Switches

A switch connects devices within the same network.

Switches commonly operate at the Data Link Layer and use MAC addresses to forward network frames.

5.3 Gateways

A gateway connects different networks or systems and can allow communication between environments that use different protocols or communication methods.

5.4 Firewalls

A firewall monitors and controls network traffic based on security rules.

It can allow legitimate traffic and block unauthorized or unwanted traffic.

5.5 Load Balancers

A load balancer distributes requests across multiple servers.

This can help improve application performance and availability.

5.6 VPN Servers

A VPN server establishes and terminates VPN connections.

It allows authorized users to create secure connections to private network resources.

6. OSI MODEL

The OSI Model is a conceptual model used to explain how network communication works.

It divides communication into seven layers:

- Physical Layer
- Data Link Layer
- Network Layer
- Transport Layer
- Session Layer
- Presentation Layer
- Application Layer

Each layer has a specific function.

6.1 Physical Layer

The Physical Layer is responsible for transmitting bits through a physical or wireless medium.

Examples include:

- Ethernet cables
- Fiber-optic cables
- Wireless signals
- Network connectors

6.2 Data Link Layer

The Data Link Layer handles communication between devices on the same local network.

It uses MAC addresses.

Switches commonly operate at this layer.

6.3 Network Layer

The Network Layer is responsible for logical addressing and routing.

IP operates at this layer.

In the practical, the source IP was: 192.168.94.128 and the destination IP was:192.168.94.134

6.4 Transport Layer

The Transport Layer provides communication between applications.

The two important protocols are:

- TCP
- UDP

It also uses port numbers to identify services.

In the practical, HTTP communication used TCP source port 57626 and destination port 80.

6.5 Session Layer

The Session Layer manages communication sessions between applications.

It helps establish, maintain, and terminate communication sessions.

6.6 Presentation Layer

The Presentation Layer deals with how data is represented.

It can be associated with functions such as:

- Data formatting
- Encoding
- Encryption
- Compression

6.7 Application Layer

The Application Layer provides network services used by applications.

Examples include:

- HTTP
- HTTPS
- DNS
- FTP
- SSH

In this practical, HTTP traffic was captured and analyzed.

7. TCP/IP MODEL

The TCP/IP model explains how communication occurs across modern networks.

The main TCP/IP layers are:

- Application Layer
- Transport Layer

- Internet Layer
- Network Access Layer

Examples of TCP/IP protocols include:

- IP
- TCP
- UDP
- HTTP
- HTTPS
- DNS
- SSH

The TCP/IP model is important because it provides the foundation for communication across modern computer networks and the Internet.

8. INTERNET PROTOCOL AND IP ADDRESSES

Internet Protocol, commonly known as IP, provides rules for addressing and routing packets.

When information is transmitted across a network, it is divided into smaller units called packets.

An IP packet contains information such as:

- Source IP address
- Destination IP address
- Packet length
- Time to Live
- Transport protocol information

An IP address identifies a device or network interface.

During the practical:

- Kali Linux IP address: 192.168.94.128
- DC-1 IP address: 192.168.94.134

Wireshark displayed these addresses when the communication between Kali Linux and DC-1 was captured.

9. TCP AND UDP

9.1 Transmission Control Protocol (TCP)

TCP is a connection-oriented transport protocol.

It is designed to provide reliable and ordered communication.

TCP provides:

- Reliable delivery
- Ordered data

- Acknowledgements
- Retransmission of missing data
- Connection establishment

In the practical, HTTP communication was transported using TCP.

The captured packet showed:

- Source port: 57626
- Destination port: 80

9.2 **User Datagram Protocol (UDP)**

UDP is another transport protocol.

Unlike TCP, UDP does not establish a connection before sending data.

UDP does not guarantee that packets will:

- Arrive
- Arrive in the correct order
- Be retransmitted if lost

UDP has lower overhead than TCP and can be useful where speed is important.

9.3 **TCP and UDP Difference**

TCP focuses on reliability, while UDP focuses on lower overhead and faster communication.

TCP is connection-oriented and provides acknowledgements and retransmissions.

UDP is connectionless and does not guarantee delivery or ordering.

10. **NETWORK PACKETS**

A network packet is a small unit of data transmitted across a network.

Packets contain information that helps the data travel from the source to the destination.

A packet can contain:

- Source address
- Destination address
- Protocol information
- Port information
- Data

Wireshark allows network professionals to examine these details.

In the practical, the selected packet contained information from several protocols:

- Ethernet

- IP
- TCP
- HTTP

This showed how different networking layers work together during communication.

11. **FIREWALLS**

A firewall is a security mechanism that monitors and controls network traffic. It can examine incoming and outgoing traffic and use security rules to determine whether traffic should be allowed or blocked.

Firewall rules can consider information such as:

- Source IP address
- Destination IP address
- Port
- Protocol
- Connection information

For example, a firewall may allow authorized HTTP or HTTPS traffic while blocking unauthorized connections. Firewalls help reduce the attack surface of a network.

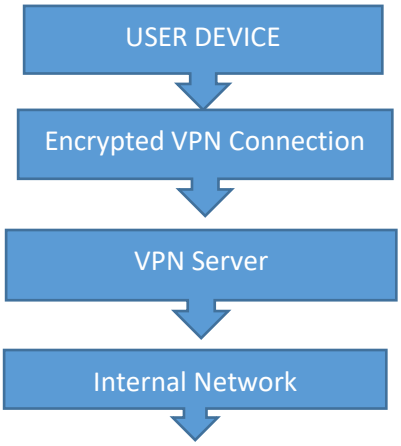
However, a firewall should not be the only security control. Organizations can also use:

- Authentication
- Access control
- Encryption
- Network monitoring
- Network segmentation
- Endpoint security

12. **VIRTUAL PRIVATE NETWORKS (VPNs)**

A Virtual Private Network, or VPN, creates a secure connection between a user and a VPN server. VPN traffic is encrypted between the user and the VPN server. VPNs are useful for users who need to securely access private organizational resources from another location.

A simple VPN communication process can be represented as:





VPNs therefore provide a way for authorized users to securely connect to private resources over an external network.

13. **NETWORK SECURITY**

Network security involves protecting networks, devices, systems, communication, and data from unauthorized access and attacks.

Network security can include:

- Firewalls
- VPNs
- Authentication
- Access control
- Encryption
- Network monitoring
- Network segmentation
- Intrusion detection
- Intrusion prevention

Network security is important because exposed services can provide information that may be useful during an attack. Tools such as Nmap and Wireshark provide different types of network visibility. Nmap can identify open ports and services, while Wireshark can provide detailed information about network traffic.

14. **NMAP**

Nmap stands for Network Mapper. It is a network scanning tool used to discover hosts, identify open ports, and identify services running on systems. Nmap can help cybersecurity professionals understand which services are exposed on an authorized system.

In this practical, Nmap was used against the intentionally vulnerable DC-1 machine from VulnHub.

The scan identified several open ports and the services associated with them.

15. **WIRESHARK**

Wireshark is a network protocol analyzer. It allows users to capture and inspect network packets.

Wireshark can display information such as:

- Source IP
- Destination IP
- Protocol
- Source port
- Destination port

- Packet length
- TCP information
- HTTP information

Wireshark can be used for:

- Network troubleshooting
- Protocol analysis
- Security investigations
- Traffic analysis
- Learning networking concepts

In this task, Wireshark was used to capture and analyze traffic generated when Kali Linux accessed the DC-1 web server.

16. **PRACTICAL ENVIRONMENT**

The practical was performed in a controlled virtual laboratory.

Analysis machine: Kali Linux

Target machine: DC-1 from VulnHub

Tools used:

- Netdiscover
- Nmap
- Wireshark
- Firefox

The main IP addresses were:

Kali Linux: 192.168.94.128

DC-1: 192.168.94.134

The practical followed a controlled process of discovering the target, scanning it, accessing an available service, and analyzing the resulting network traffic.

17. **PRACTICAL METHODOLOGY**

The practical was completed using the following steps:

- Started the Kali Linux and DC-1 virtual machines.
- Used Netdiscover to identify devices on the network.
- Identified DC-1 at 192.168.94.134.
- Used Nmap to scan the DC-1 machine.
- Identified open ports and services.
- Identified HTTP running on port 80.
- Entered 192.168.94.134 into Firefox.

- Accessed the DC-1 web server.
- Started Wireshark to capture the network traffic.
- Selected a relevant HTTP packet.
- Analyzed the source and destination IP addresses.
- Analyzed TCP port information.
- Analyzed the HTTP request.
- Connected the practical results to networking concepts.

18. **NETWORK DISCOVERY USING NETDISCOVER**

The first practical step was to discover devices on the local network.

The following command was used: **netdiscover**

The output showed several devices, including:

- 192.168.94.1
- 192.168.94.2
- 192.168.94.134
- 192.168.94.254

The DC-1 target was identified at:

192.168.94.134

This IP address was then used for the Nmap scan.

19. **NMAP SCANNING**

After identifying the DC-1 IP address, Nmap was used to scan the target.

The command used was: ***nmap -P -sV -O 192.168.94.134***

The scan showed that the host was active.

The following open TCP ports were identified:

- **22/tcp open ssh**
- **80/tcp open http**
- **111/tcp open rpcbind**

The identified services were:

- Port 22: SSH
- Service: OpenSSH 6.0p1 Debian 4+deb7u7
- Port 80: HTTP
- Service: Apache httpd 2.2.22 (Debian)
- Port 111: RPCbind
- Service:
- rpcbind 2-4

Nmap also identified the target as a Linux-based system.

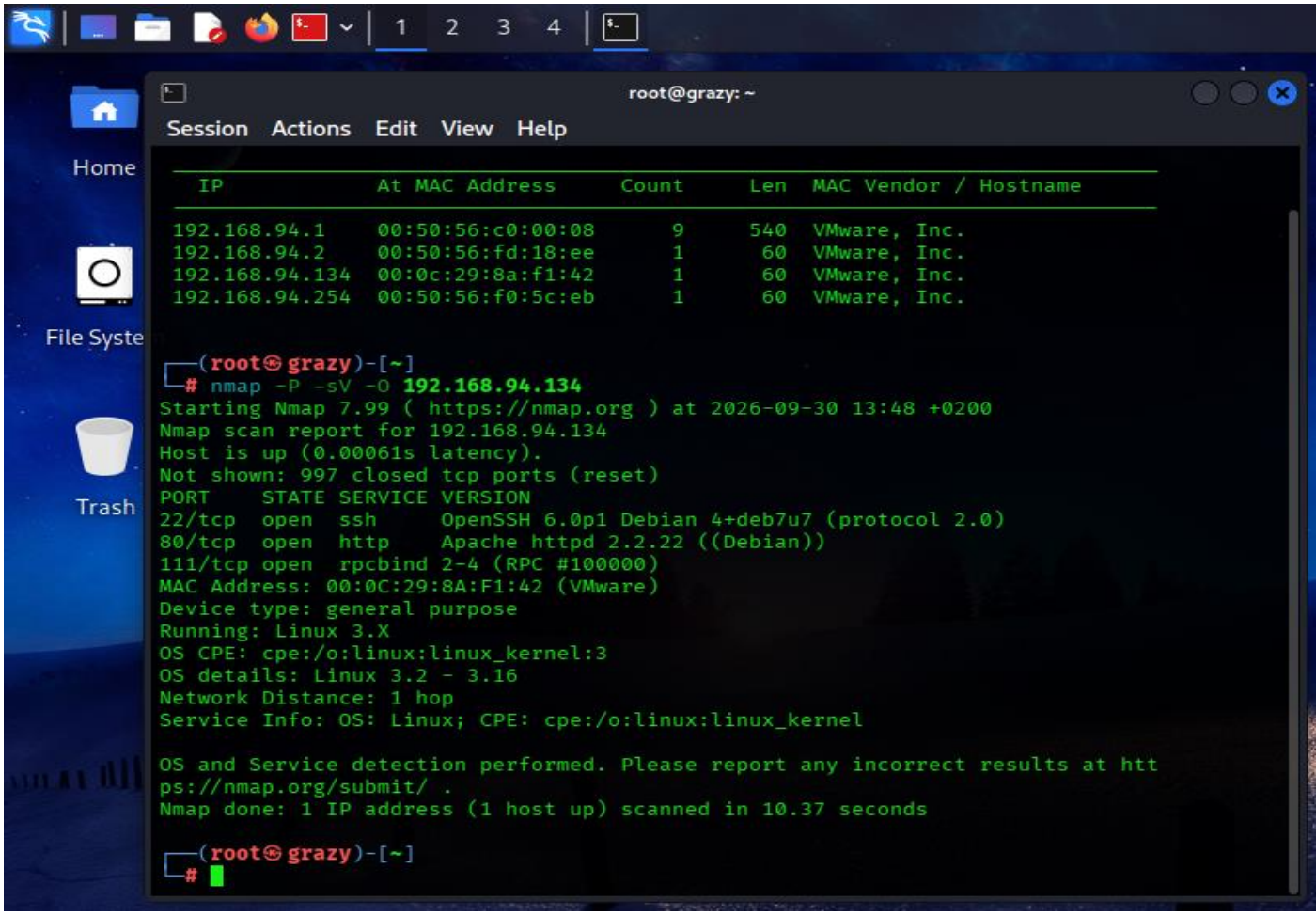


Figure 1: Nmap scan results for the DC-1 target

20. **ANALYSIS OF NMAP RESULTS**

The Nmap results showed that the DC-1 machine had several network services available.

- **Port 22**

Port 22 was open and running SSH. SSH is commonly used for secure remote administration.

- **Port 80**

Port 80 was open and running HTTP. The web server was identified as: Apache httpd 2.2.22 (Debian). This was the service used for the next stage of the practical.

- **Port 111**

Port 111 was open and running RPCbind. This demonstrated that the target had multiple network services exposed.

The Nmap scan therefore provided an overview of the services available on the target before network traffic was analyzed using Wireshark.

21. **ACCESSING THE DC-1 WEB SERVER**

After identifying HTTP on port 80, the target IP address was entered into Firefox:

http://192.168.94.134

The browser connected to the DC-1 web server. This generated network traffic between the Kali Linux machine and DC-1.

Wireshark was then used to capture this communication.

22. **WIRESHARK PACKET CAPTURE**

Wireshark was opened on Kali Linux and the appropriate network interface was selected. Packet capture was started before accessing the DC-1 web server. When Firefox accessed: **http://192.168.94.134** network traffic was generated.

The Wireshark capture showed:

- Packets: 490
- Dropped: 0.0%

This means that 490 packets were captured and Wireshark reported no dropped packets during the capture.

The captured traffic included communication between:

- Source: 192.168.94.128
- Destination: 192.168.94.134

23. **WIRESHARK PACKET ANALYSIS**

A relevant packet was selected for detailed analysis. The selected packet was Frame 455.

The packet showed:

- Source IP: 192.168.94.128
- Destination IP: 192.168.94.134
- Protocol: HTTP
- Source port: 57626
- Destination port: 80
- The packet contained the following HTTP request: GET /themes/bartik/images/buttons.png HTTP/1.1

This shows that the browser requested an image file from the DC-1 web server.

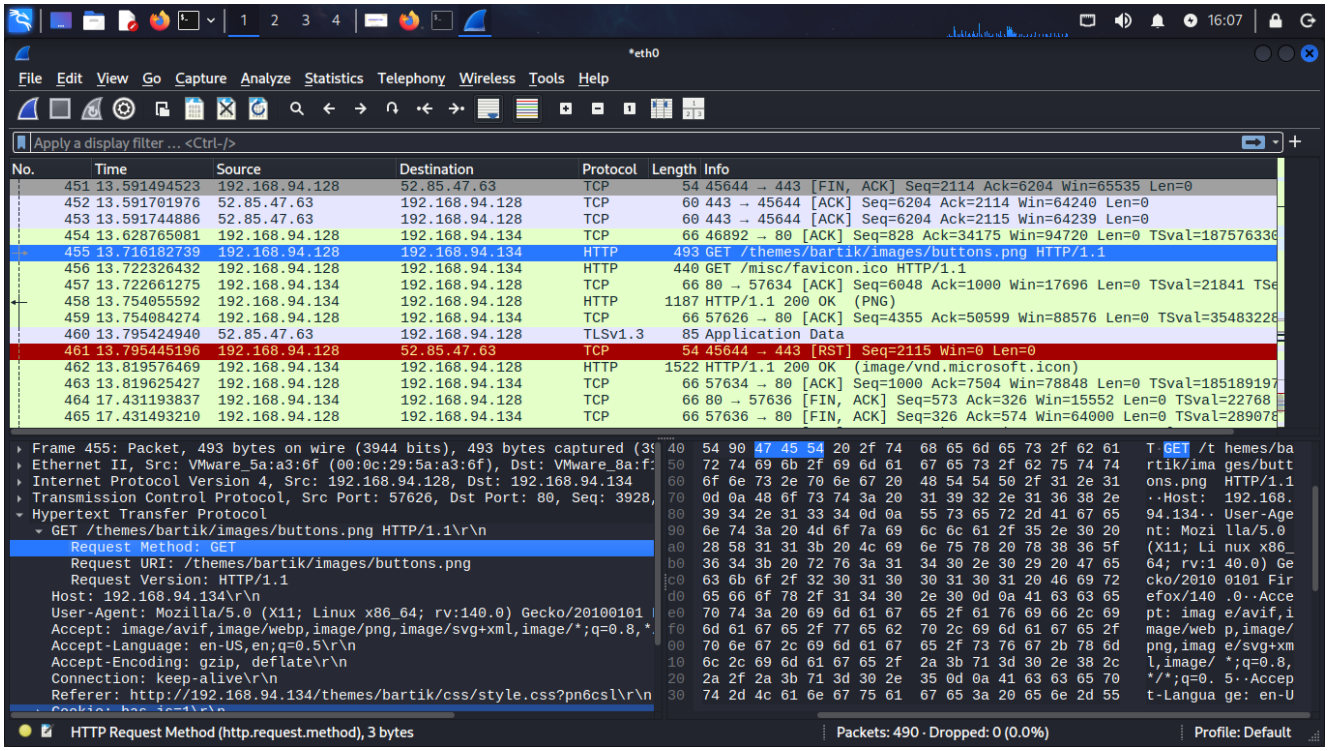


Figure 2: Wireshark capture showing HTTP traffic between Kali Linux and DC-1

24. **DETAILED ANALYSIS OF THE HTTP REQUEST**

The selected HTTP packet provided several useful details.

- **HTTP Request Method**

The request method was: GET

The GET method is used by a client to request a resource from a web server.

- **Requested Resource**

The requested resource was: ***/themes/bartik/images/buttons.png***

This shows that the browser requested an image file from the server.

- **HTTP Version**

The request used: HTTP/1.1

- **Host**

The packet showed:

Host: 192.168.94.134

This identifies the web server being contacted.

- **User-Agent**

The packet also contained User-Agent information. It showed that the request came from Firefox running on Linux x86-64. This demonstrates that HTTP headers can provide information about the client making a request.

- **TCP Information**

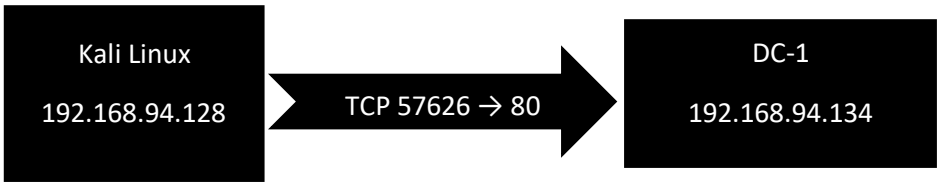
The HTTP request was carried using TCP.

The packet showed:

Source port: 57626

Destination port: 80

Therefore, the communication was:



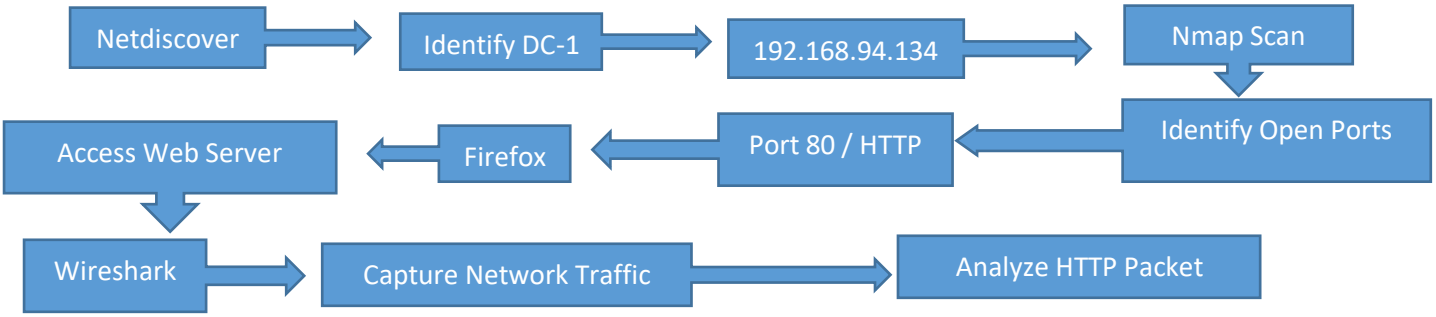
25. RELATIONSHIP BETWEEN NMAP AND WIRESHARK

Nmap and Wireshark performed different but related functions during the practical.

Nmap was used to identify open ports and services.

Wireshark was used to examine actual network traffic.

The practical workflow was:



This demonstrated how different networking tools can be used together for network analysis.

26. SECURITY IMPORTANCE OF THE PRACTICAL

The practical demonstrated why network visibility is important in cybersecurity.

Nmap showed which services were exposed on the DC-1 system.

The scan identified:

- SSH
- HTTP
- RPCbind

Knowing which services are running helps security professionals understand the network attack surface of a system.

Wireshark provided more detailed information about the actual communication.

It allowed the traffic to be examined using:

- Source IP
- Destination IP
- Source port
- Destination port
- Protocol
- HTTP method
- Requested resource
- Host information
- User-Agent information

This type of network visibility can help with troubleshooting, monitoring, and security investigations.

27. K**EY LESSONS LEARNED**

The practical and theoretical work provided the following lessons:

- Computer networks allow devices to communicate and share information.
- The OSI Model provides a structured way of understanding network communication.
- The TCP/IP model explains important protocols used by modern networks.
- IP addresses identify the source and destination of network traffic.
- Ports identify network services.
- TCP provides reliable and ordered communication.
- UDP provides communication with lower overhead but does not guarantee delivery.
- Routers connect different networks.
- Switches connect devices within a network.
- Firewalls control network traffic based on security rules.
- VPNs provide secure encrypted connections for authorized users.
- Nmap can identify open ports and services.
- Wireshark can capture and analyze network packets.
- Network packets contain information about communication between systems.
- Nmap and Wireshark can be used together during network analysis.
- Practical network analysis helps connect theoretical networking concepts to real network communication.

28. **CHALLENGES AND OBSERVATIONS**

One important observation from the practical was that the DC-1 machine had multiple services exposed.

The Nmap scan identified:

- 22/tcp SSH
- 80/tcp HTTP
- 111/tcp RPCbind

This showed that a single system can provide multiple network services.

Another observation was that accessing a web server generates many network packets.

Wireshark captured 490 packets during the practical.

The selected HTTP packet showed:

- ***GET /themes/bartik/images/buttons.png HTTP/1.1***

This demonstrated that loading a web page involves multiple requests between the browser and web server.

The practical also demonstrated a direct relationship between Nmap and Wireshark.

Nmap identified HTTP on port 80, and Wireshark later showed HTTP communication using destination port 80.

29. **PRACTICAL RESULTS SUMMARY**

ITEM	**Result**
Operating System	Kali Linux
Target	DC-1 from VulnHub
Kali Linux IP	192.168.94.128
DC-1 IP	192.168.94.134
Port 22	Open
Service	SSH
Port 80	Open
Service	HTTP
Web Server	Apache httpd 2.2.22 (Debian)
Port 111	Open
Service	RPCbind
Packets Captured	490
Dropped Packets	0.0%
Selected Frame	455
Protocol	HTTP
Source Port	57626
Destination Port	80
HTTP Method	GET
Requested Resource	/themes/bartik/images/buttons.png

30. **CONCLUSION**

Networking is an important foundation of cybersecurity because computers, servers, applications, and users depend on networks to communicate.

This task provided an understanding of important networking concepts, including the OSI Model, TCP/IP, IP addresses, network packets, TCP, UDP, network devices, firewalls, VPNs, Nmap, and Wireshark.

The practical exercise was performed using Kali Linux and the intentionally vulnerable DC-1 virtual machine from VulnHub.

Netdiscover was first used to identify the target machine. DC-1 was identified at:

- 192.168.94.134

Nmap was then used to scan the target. The scan identified three open services:

- 22/tcp SSH
- 80/tcp HTTP
- 111/tcp RPCbind

Port 80 was identified as an HTTP service running Apache httpd 2.2.22 on Debian.

The IP address of DC-1 was then entered into Firefox to access the web server.

Wireshark was used to capture and analyze the resulting network traffic. A total of 490 packets were captured, with 0.0% reported as dropped.

Frame 455 was selected for detailed analysis. It showed communication between:

- Source: 192.168.94.128
- Destination: 192.168.94.134

The packet used TCP source port 57626 and destination port 80.

The HTTP request was:

GET /themes/bartik/images/buttons.png HTTP/1.1

This practical demonstrated how networking concepts can be observed in a real network environment. It also showed the relationship between network discovery, service identification, and packet analysis.

Overall, the task provided practical experience with network discovery using Netdiscover, service scanning using Nmap, and packet capture and analysis using Wireshark. These skills are important for understanding network communication and performing cybersecurity-related network analysis.